

TRAFFIC LIGHT SIMULATOR

■ BUKYA SRIDHAR

Traffic lights display three colour lights cyclically to control the traffic:

1. Red to 'stop' the traffic from moving
2. Yellow signals traffic to 'wait'
3. Green allows it to 'go' ahead

The cycle is automatic and repeats periodically. In this traffic light simulation program, red light is displayed for 25 seconds, yellow for 10 seconds and green for 15 seconds, and the cycle repeats automatically. The program has been developed using VML (Vector Markup Language), XML (Extensible Markup Language) and JavaScript.

VML is used to draw the vector diagram (in this case, the circle), XML for the timing information stored in the database and JavaScript for the main program. The program code is very small and easy to understand.

Program execution

When the program (Traffic Light.htm) is run in Internet Explorer, a timer counts 24 seconds down to 0. Simultaneously, it displays a red circle indicating 'stop,' which can be seen on the right-hand-side corner of the screen. When the count reaches 0, the timer restarts counting from 9 and the



Fig. 1: Screenshot showing red active



Fig. 2: Screenshot showing yellow active



Fig. 3: Screenshot showing green active

circle turns yellow indicating 'wait.' When the count reaches 0, the circle turns green for 15 seconds. Thereafter, the cycle repeats automatically.

In order to make the application

more attractive and the screen informative, three independent circles have been programmed to show the change in colour lights while the countdown timer counts.

Function description

This program uses five functions. Traffic() function scans the XML database for the task parameter and tells the program what is to be done. Timer() function scans the database for time in decimal format and converts it into sexagesimal system—the degrees, minutes and seconds format used in clocks. The program takes a decimal value for time. This value needs to be converted into clock time format in order to use it as a clock for countdown. Countdown() function counts the converted time down to 0 second. Halt() function stops the counter when the cycle is complete, and the program calls traffic() function again to repeat the cycle from start.

EFY note. The source codes of this article have been included in this month's EFY-CD and are also available on the www.efymag.com website.



TRAFFIC LIGHT.HTM

```
<html xmlns:t>
<head>
<xml id="T" src="T.xml"></xml>
<title>Traffic Island Simulator</title>
<style>t:*(behavior:url(#default#VML));</style>
<style>
.T
{
border:1px solid black;
font-family:Neuropol;
font-size: 20pt;
filter:progid:DXImageTransform.Microsoft.Gradient(startColorstr=#f0f0f0,endColorstr=#d0d0d0);
Type=1;
}
```

```
color:Yellow;
cursor:Text;
}
</style>
<noscript style="font-family:Neuropol">This Application Requires JavaScript Plug-in which has to be initialized or installed. It seems that your OS does not have the Plug-in Installed or has been disabled by you. Please Enable/Install the Plug-in. If you have the plug-in and is currently disabled, right-click on the light yellow bar above, Chose "Allow Blocked Content..." and then click "Yes" from the consecutive dialog box, you should
```

```
see the Application running, or else install the plug-in from Internet/Reliable Disc.</noscript>
<script language="JavaScript">
Signal=T.recordset;
function Traffic(form)
{
document.getElementById("Task").innerHTML=Signal("msg");
LED();
if (Signal.EOF) {Signal.moveFirst();Traffic(this.form)}
Halt(this.form);
Timer(this.form);
}
function Timer(form)
```